

JILL WANG

Data Scientist & AI/ML Engineer

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WORK EXPERIENCE

Head of Analytics

Jul 2023 – Present

Truthifi – digital finance monitoring platform · Princeton, NJ

- **MCP Server & LLM Application Development:**
 - Built and shipped 25+ MCP tools serving an internal AI agent and external third-party MCP clients, powering a complete, customizable financial health analysis experience
 - Built rich metadata contracts for third-party MCP clients, user friendly tool inputs, MongoDB-backed pagination to optimize query load for large result sets, and MongoDB-backed vector search for answering help questions
 - Built and shipped a browsable prompt catalog that surfaces curated financial analysis workflows to users
 - Engineered an LLM application with Langgraph to build custom agent workflow, and produced prompt safety guardrails
- **Conversion & Growth Analytics:**
 - Shipped directive MCP tools and tier-based upgrade nudges that drove account-linking conversion, subscription growth, and rising assets under monitoring
 - Unified the upgrade experience across subscription tiers, combining PostHog hooks, React hooks, and IntersectionObserver to register impressions versus clicks on upgrade call-to-action components, giving accurate engagement analytics
 - Built SQL-driven PostHog dashboards and funnel analyses for the marketing public site, core web app, MCP server, and internal agent
- **Data Engineer & Analytics Development:**
 - Built efficient ETL pipelines for vendor APIs (e.g. Xignite, Morningstar, Yahoo Finance) and government sources (e.g. SEC, MSRB) that meticulously standardize, validate, and enhance millions of records to maintain the quality of the securities master database
 - Developed real-time performance feedback and diagnostic metrics, portfolio composition breakdown, and benchmarks for user's investment portfolios
- *Technology: TypeScript, MongoDB, Model Context Protocol (MCP), React, Langchain, Node.js, NestJS, GraphQL*

Data Scientist (via Adastra)

Mar 2023 – Jun 2023

Mark Anthony Group – alcoholic beverage company · Toronto, Canada

- Built an XGBRegressor sales-forecasting ML model on PySpark pipelines processing millions of rows to power gap and attention marketing analysis
- *Technology: Python, AWS SageMaker, XGBoost, SQL*

Operation Research Analyst (via Adastra)

Aug 2022 – Feb 2023

Taiga Building Products Ltd. – wholesale building materials distributor · Toronto, Canada

- Built an address-clustering AI model (Google Direction API) and a truck-load optimization model by mathematical formulation (Google OR-Tools Linear Programming) to optimize delivery routing and truck-loads, reducing freight costs and improving sales margins
- *Technology: Python, Azure ML, Google OR-Tools, Google Direction API*

Data Analytics Developer (via Adastra)

May 2021 – Jun 2022

TD Bank – project delivery team · Toronto, Canada

- **Security Classification & Retention Period Prediction Model:**
 - Owned the end-to-end data science lifecycle for the model (data lifecycle management application), partnering with data stewards to gather and validate training data across company digital assets
 - Engineered text features using vector embeddings and NLP techniques to capture semantic signals from document content, cleaned and consolidated additional structured features, then applied Principal Component Analysis (PCA) to denoise the combined feature set before training

- Trained and tuned an XGBClassifier with Hyperopt to predict security classifications and retention periods, delivering predictions in minutes that previously took months of manual work from data stewards
- **ETL & Business Intelligence Dashboards:**
 - Built ETL pipelines on the Apache big data ecosystem feeding real-time data straight from source systems into 50+ interactive Tableau executive dashboards, engineering custom formulas to handle the underlying calculations
- *Technology: Impala, Python, scikit-learn, XGBoost, Hyperopt, PCA, NLP, spaCy, tkinter, Tableau, Oozie*

Data Science Consultant

Apr 2021 – Jun 2023

Adastra North America – data and AI consultancy · Toronto, Canada

- Delivered scalable analytics and machine learning solutions across enterprise engagements at TD Bank, Taiga Building Products, and Mark Anthony Group

Data Scientist & Software Developer (Intern)

Jun 2019 – Aug 2019

Graphen – R&D group developing an anti-money-laundering platform · Taipei, Taiwan

- Developed an unsupervised model on time-series data and a supervised model on multivariate data for financial behaviour anomaly/fraud detection, applying statistical methods to design a systematic risk (anomaly) score and a numerical analysis method (power spectral density) to improve time series model accuracy
- Developed the front end of a React-based anti-money-laundering platform
- *Technology: Python, NumPy, pandas, scikit-learn, XGBoost, JavaScript, React*

SKILLS

AI Engineering: MCP · LangChain · LangSmith · OpenAI API

Data Science: Python · NumPy & pandas · scikit-learn · XGBoost · TensorFlow/Keras · spaCy · statsmodels · SciPy · R · MATLAB

ETL: MongoDB · SQL · Hadoop · Impala · Hive · Oozie

Business Intelligence: Tableau · PostHog

Software Development: TypeScript · React · JavaScript · Node.js · NestJS · GraphQL

Others: Git · Bash · C/C++

EDUCATION

Honours Bachelor of Mathematics in Computational Mathematics

University of Waterloo, 2021

Honours Bachelor of Mathematics in Statistics, Computing Technology Option

University of Waterloo, 2021

Relevant Coursework: Statistical Learning & Classification, Generalized Linear Models & Applications, Applied Probability, Computational Statistics and Data Analysis, Mathematical Statistics, Data Structures & Algorithms, Object-Oriented Software Development, Applied Cryptography, Computational Methods for Differential Equations